

## Appendix: Teacher-Authored AI Tutors at Classroom Scale: When Students Reach the Targeted Depth of Knowledge and How Long They Hold It

### 1. System Context and Sample Construction

The platform composes three instruction layers, a base commercial LLM, a fixed platform system prompt establishing safety constraints and Socratic defaults, and the teacher-authored prompt specifying role, scaffolding moves, constraints, task objective, and completion criterion for a specific activity.

Of 1,479 conversations generated during the deployment, 117 were excluded from the conversation-level dataset. Thirty came from three activities whose prompts did not support a target DOK assignment. For 87, the coding pipeline returned no demonstrated-DOK label, in nearly all cases from JSON encoding failures on verbatim evidence spans rather than from uncodable student content; coding these conversations with a corrected parser and adding them back left the central quantities unchanged. Turn-level coding covered every turn of the 1,213 primary-sample conversations (23,114 turns), so the primary sample carries complete turn coverage.

Student turn 1 in every conversation is the platform's opening button ("Start Classroom Discussion"). It carries no preceding AI move and cannot demonstrate DOK. All turn indices in the analyses therefore count typed student turns, the hazard risk set begins at the first typed turn, and the 84 conversations (6.9%) containing only the button are retained as not achieved. Excluding them raises DOK 3 achievement from 47.1% to 51.0%.

### 2. Turn-Level Codebook

**Student turns.** Each student turn receives one DOK level. Level 0 marks procedural or non-substantive turns, including platform boilerplate. Level 1 marks recall, factual answers, and definitions. Level 2 marks explanation, application, and comparison in the student's own words. Level 3 marks strategic thinking, argument with evidence, synthesis, and evaluation of alternatives. Level 4 marks extended thinking sustained across frames. Coding uses only what the student writes, with AI turns as context.

**AI turns.** Each AI turn receives one or more move codes. Socratic question, an open question that asks the student to reason. Stepwise guidance, an ordered next step. Hint, partial information short of the answer. Request attempt, asking the student to try before help. Request evidence, asking for a source, example, or reason. Affirm and extend, accepting a student idea and pushing it further. Final answer, providing the deliverable or answer. Redirect, returning an off-task turn to the task. Recap, summarizing. Closure, signaling the finish line. Other.

**Primary move for sequence mining.** Multi-coded AI turns are collapsed to one primary move by fixed priority, final answer, redirect, request evidence, request attempt, stepwise guidance, hint, Socratic question, affirm and extend, recap, closure, other. The priority ranks rarer, higher-commitment moves above frequent defaults so that sequence mining does not reduce every turn to the modal code.

### 3. Coding Instruments

The complete instruments given to the LLM coder for each coding layer are reproduced verbatim below. Placeholders in braces are filled per unit at run time. The instruments, together with the coded datasets and the analysis code, are available in the replication package.

#### C.1 Layer 1. Prompt-coding instrument

Applied to each teacher-authored prompt; output was reviewed to full human consensus.

SYSTEM

You are an expert learning-sciences researcher and classroom assessment specialist. Your job is to code teacher-authored AI \setup prompts" (discussionContent) into a fixed rubric of instructional features and a target Depth of Knowledge (DOK) level. You must be conservative, literal, and evidence-based: only code a feature as present if it is explicitly supported by the prompt text. Do NOT infer teacher intent beyond what is written.

You MUST output valid JSON only (no markdown, no commentary, no extra keys). If a field is unknown or not evidenced, use the specified default (\unknown", null, or 0). Use the coding rules and definitions below.

USER

You will be given a single teacher-authored setup prompt (discussionContent). Code it using the rubric.

## CODING RUBRIC (discussion-level)

## A) Task specificity &amp; finish line

- task\_specificity:
  - 0 = vague (goal unclear; no concrete product)
  - 1 = moderate (general task stated; limited success criteria)
  - 2 = high (clear deliverable and/or steps and/or success criteria)
- finish\_line:
  - 0 = no explicit endpoint/deliverable
  - 1 = explicit deliverable and/or completion criterion (e.g., \produce X", \stop after Y", \when done, ...")

## B) Scaffolding strategy (mark all that apply; 0/1 each)

- scaffold\_socratic: prompt instructs AI to ask questions that probe student thinking rather than provide answers
- scaffold\_stepwise: prompt instructs multi-step process, sequencing, or gradual progression
- scaffold\_hints\_first: prompt says provide hints/clues before full explanations/solutions
- scaffold\_attempt\_first: prompt requires student attempt before AI helps further
- scaffold\_worked\_example\_prohibition: prompt explicitly forbids giving full solutions/finished work (e.g., \do not give the answer", \don't write it for them")

## C) Epistemic framing (0/1 each; mark if explicitly required)

- epistemic\_explain\_reasoning: requires explaining reasoning/steps/why
- epistemic\_use\_evidence: requires citing evidence, quoting text, referencing sources, or \use evidence"
- epistemic\_compare\_alternatives: requires comparing, contrasting, evaluating options/strategies
- epistemic\_justify\_claims: requires justification/argumentation (claims + support)

## D) Constraints (0/1 each; mark if explicitly stated)

- constraint\_short\_turn: limits length (e.g., \1{2 sentences", \brief", \short")
- constraint\_one\_question\_per\_turn: \one question at a time" or similar
- constraint\_structured\_format: requires a format/template (bullets, table, CER, steps, rubric)
- constraint\_language\_level: mentions reading level, age-appropriate language, ELL supports, or vocabulary constraints

## E) Role and tone

- role: one of ["tutor", "coach", "interviewer", "peer", "critic", "facilitator", "teacher", "other", "unknown"]  
Choose the closest role explicitly stated (or strongly implied by explicit instructions like \ask me questions" -> interviewer).
- tone: one of ["supportive", "neutral", "strict", "playful", "unknown"]  
Only code if tone is explicitly described (e.g., \encouraging", \firm", \fun").

## F) Safety/ethics guardrails (0/1 each; only if explicit)

- guardrail\_no\_direct\_answers: says not to provide direct answers/solutions
- guardrail\_integrity\_language: mentions cheating, plagiarism, academic honesty, \don't copy"
- guardrail\_privacy: mentions personal data, privacy, or safety rules
- guardrail\_respectful: mentions respectful language, harm, bias, or appropriateness

## G) Prompt target DOK (prompt\_target\_dok)

Assign the DOK level required by the prompt's objective (not what the student might do in practice).

- 1 = Recall / reproduce: facts, definitions, simple retrieval, listing
- 2 = Skills / concepts: explain concepts, summarize, apply procedure, classify, organize
- 3 = Strategic thinking: justify reasoning, analyze, compare, use evidence, critique, revise with rationale
- 4 = Extended thinking: synthesize across sources/time, design investigation, multi-stage project, sustained reasoning over time

If multiple tasks exist, choose the highest DOK that is essential to complete the stated deliverable. If DOK is ambiguous, choose the lower plausible level and set dok\_uncertainty = true.

## OUTPUT FORMAT (JSON only)

Return a single JSON object with exactly these keys:

```
{
```

```

"task_specificity": 0|1|2,
"finish_line": 0|1,

"scaffold_socratic": 0|1,
"scaffold_stepwise": 0|1,
"scaffold_hints_first": 0|1,
"scaffold_attempt_first": 0|1,
"scaffold_worked_example_prohibition": 0|1,

"epistemic_explain_reasoning": 0|1,
"epistemic_use_evidence": 0|1,
"epistemic_compare_alternatives": 0|1,
"epistemic_justify_claims": 0|1,

"constraint_short_turn": 0|1,
"constraint_one_question_per_turn": 0|1,
"constraint_structured_format": 0|1,
"constraint_language_level": 0|1,

"role":
"tutor"|"coach"|"interviewer"|"peer"|"critic"|"facilitator"|"teacher"|"other"|"unknown",
"tone": "supportive"|"neutral"|"strict"|"playful"|"unknown",

"guardrail_no_direct_answers": 0|1,
"guardrail_integrity_language": 0|1,
"guardrail_privacy": 0|1,
"guardrail_respectful": 0|1,

"prompt_target_dok": 1|2|3|4,
"dok_uncertainty": true|false,

"evidence_spans": {
  "finish_line": [<verbatim snippets>],
  "scaffolding": [<verbatim snippets>],
  "epistemic": [<verbatim snippets>],
  "constraints": [<verbatim snippets>],
  "role_tone": [<verbatim snippets>],
  "guardrails": [<verbatim snippets>],
  "dok": [<verbatim snippets>]
}
}

```

#### EVIDENCE SPANS RULES

- Each list should contain 0{3 short verbatim snippets (max ~15 words each) taken directly from discussionContent that justify your codes.
- If a category has no direct evidence, output an empty list for that category.
- Do NOT quote more than 15 words per snippet. Do NOT include any student names or personal data if present.

Now code the following discussionContent:

<DISCUSSION\_CONTENT\_HERE>

## C.2 Layer 2. Conversation-level student-DOK instrument

Applied to each conversation; produces the conversation-level demonstrated-DOK label.

You are an expert learning-sciences coder. You will assess the Depth of Knowledge (DOK) demonstrated by a STUDENT in a conversation with an AI tutor.

**CRITICAL INSTRUCTION:** You must code DOK based ONLY on what the student writes in their messages. The AI responses provide context for understanding what the student is responding to, but the DOK level reflects the cognitive demand demonstrated in the STUDENT'S OWN WORDS | their requests, explanations, arguments, and reasoning.

Do NOT code DOK based on:

- The complexity of the AI's responses
- The sophistication of the AI's questions
- What the AI says the student should do
- The cognitive demand of the task as described by the teacher

DO code DOK based on:

- The student's actual written output
- Whether the student recalls, explains, analyzes, or synthesizes
- The reasoning visible in student turns
- The evidence, argumentation, or strategic thinking the student produces

DOK Level Definitions (applied to STUDENT messages only):

#### DOK 1 | Recall / Reproduce

Student responses are limited to: listing facts, one-word or short answers, yes/no responses, copying/repeating information, simple identification, basic definitions. The student retrieves or states information without elaboration.

#### DOK 2 | Skills / Concepts

The student: explains a concept in their own words, summarizes information, applies a known procedure, describes relationships between ideas, classifies or organizes, compares at a surface level. The student engages with content but does not construct original analysis or argumentation.

#### DOK 3 | Strategic Thinking

The student: constructs an argument with evidence, justifies their reasoning ("because..." with substantive rationale), analyzes by breaking down and evaluating, synthesizes across multiple sources or perspectives, compares using explicit criteria, critiques with reasoning, revises their own thinking with stated rationale. The student produces original analytical reasoning visible in their text.

#### DOK 4 | Extended Thinking

The student: integrates multiple frameworks or perspectives over sustained turns, proposes novel connections between domains, designs an investigation or solution approach, engages in multi-step reasoning that builds across many turns. This level requires sustained, cumulative intellectual work across the conversation.

#### DECISION RULES:

- If a student asks a factual question ("what is X?"), that is DOK 1 regardless of how complex the topic is.
- If a student explains their understanding ("I think X works because Y"), that is DOK 2.
- If a student argues with evidence ("The evidence shows X, which means Y, and this contradicts Z because..."), that is DOK 3.
- If ambiguous between two levels, choose the LOWER level.
- A single sentence of reasoning surrounded by recall-level messages means DOK max = the level of that best sentence.

Output valid JSON only. No markdown, no commentary.

--- User message template ---

Assess the student's demonstrated DOK level in this conversation.

CONTEXT (for understanding only | do NOT base DOK on these):

- Teacher's setup prompt: {teacher\_prompt}
- Target DOK the teacher intended: {target\_dok}

STUDENT MESSAGES TO CODE (assess DOK from these):

{student\_messages}

AI MESSAGES (context only | shows what the student was responding to):

{ai\_messages}

Code the following JSON:

```
{
  "student_dok_max": <1|2|3|4>,
  "student_dok_final": <1|2|3|4|null>,
  "confidence": "<high|medium|low>",
  "rationale": "<one sentence explaining why you chose this DOK level, citing the specific student text>",
  "evidence_max": "<verbatim quote from the student message that best demonstrates the max DOK level, max 30 words>",
  "evidence_final": "<verbatim quote from the student's final message showing DOK level, max 30 words>"
}
```

Confidence definitions:

- high: the student text clearly and unambiguously demonstrates this DOK level with strong evidence
- medium: the DOK level is the best fit but could reasonably be coded one level higher or lower
- low: ambiguous or borderline case where the evidence is weak or mixed

### C.3 Layer 3. Turn-level coding instrument

Applied to every turn of every primary-sample conversation, in windows of 40 turns with 4 preceding turns shown as context.

You are an expert learning-sciences coder. You code each turn of a student-AI conversation that took place under a teacher-authored activity prompt.

You will code TWO kinds of turns.

1. AI turns. Assign every AI-move code that the turn clearly exhibits, from this list:

["socratic\_question", "stepwise\_guidance", "hint", "request\_attempt", "request\_evidence", "affirm\_extend", "final\_answer", "redirect", "recap", "closure", "other"]

Definitions:

- socratic\_question: an open question that asks the student to reason, explain, predict, or evaluate (not a yes/no or factual check).
- stepwise\_guidance: breaks the task into an explicit next step or an ordered sequence of steps.
- hint: partial information or a nudge toward the answer that stops short of giving it.
- request\_attempt: asks the student to try, draft, or answer before further help is given.
- request\_evidence: asks for a source, example, quotation, data point, or a reason for a claim.
- affirm\_extend: accepts or praises a student idea AND pushes it further (a follow-up that builds on what the student said).
- final\_answer: gives the answer, the solution, or the deliverable itself (a complete paragraph the student was asked to write, the solved problem, the direct factual answer to the task question).
- redirect: returns an off-task or shortcut request back to the task or the rules.
- recap: summarizes what has been established so far in the conversation.
- closure: signals that the task, deliverable, or finish line is complete.
- other: greeting, logistics, or generic encouragement with none of the above.

Use "other" only when no other code applies. Most substantive AI turns carry 1 to 3 codes.

2. Student turns. Assign (a) the DOK level demonstrated IN THAT TURN and (b) every student-move code that applies.

DOK levels, based ONLY on what the student writes in that turn:

- 0: non-substantive (empty, "Start Classroom Discussion", "ok", "hi", a single word with no content, a bare request).
  - 1: recall or reproduce (facts, one-word or short answers, yes/no, copying, simple identification, a factual question).
  - 2: skills and concepts (explains a concept in own words, summarizes, applies a procedure, describes a relationship, surface comparison).
  - 3: strategic thinking (constructs an argument with evidence, justifies with substantive rationale, analyzes, synthesizes sources or perspectives, compares with explicit criteria, critiques, revises with stated rationale).
  - 4: extended thinking (integrates multiple frameworks over sustained reasoning, designs an investigation or solution approach, multi-step reasoning across many ideas within the turn).
- Decision rules: a factual question is DOK 1 regardless of topic; "I think X because Y" with a substantive Y is DOK 3, with a trivial Y is DOK 2; if ambiguous choose the LOWER level; do not raise the DOK because the AI's question was demanding.

Student-move codes:

["asks\_question", "explains", "argues\_with\_evidence", "compares\_evaluates", "submits\_work", "requests\_answer", "requests\_help", "agrees\_acknowledges", "off\_task", "other"]

Output valid JSON only, no markdown, no commentary. Code ONLY the turns listed under TURNS TO CODE. Turns under CONTEXT are for understanding and must not appear in the output.

--- User message template ---

Teacher's activity prompt (context only):

{teacher\_prompt}

Target DOK the teacher intended (context only, do not use it to raise student DOK):

{target\_dok}

CONTEXT (preceding turns, do not code):  
{context\_turns}

TURNS TO CODE:  
{turns}

Return JSON of the form:

```
{
  "turns": [
    {
      "turn": <int>,
      "role": "ai",
      "codes": [<ai move codes>]
    },
    {
      "turn": <int>,
      "role": "student",
      "dok": <0|1|2|3|4>,
      "moves": [<student move codes>],
      "evidence": "<up to 12 words quoted from the turn that justify the DOK>"
    }
  ]
}
```

Include exactly one object per turn listed under TURNS TO CODE, in order.

## 4. Prompt Feature Prevalence

**Table 1.** Prompt feature prevalence across the 74 activities with codable target DOK.

| Feature                       | <i>N</i> | %     |
|-------------------------------|----------|-------|
| <b>Task structure</b>         |          |       |
| High specificity (level 2)    | 60       | 81.1% |
| Finish line present           | 51       | 68.9% |
| <b>Target DOK</b>             |          |       |
| DOK 1 (Recall)                | 4        | 5.4%  |
| DOK 2 (Skills/Concepts)       | 30       | 40.5% |
| DOK 3 (Strategic Thinking)    | 40       | 54.1% |
| <b>Scaffolding strategies</b> |          |       |
| Stepwise guidance             | 36       | 48.6% |
| Socratic questioning          | 25       | 33.8% |
| Worked example prohibition    | 12       | 16.2% |
| <b>Epistemic framing</b>      |          |       |
| Explain reasoning             | 23       | 31.1% |
| Compare alternatives          | 18       | 24.3% |
| Use evidence                  | 14       | 18.9% |
| <b>Guardrails</b>             |          |       |
| No direct answers             | 12       | 16.2% |
| Respectful tone               | 9        | 12.2% |

|              |             |                      |                   |                    |                   |              |                      |                |             |                       |                   |                   |                 |                       |
|--------------|-------------|----------------------|-------------------|--------------------|-------------------|--------------|----------------------|----------------|-------------|-----------------------|-------------------|-------------------|-----------------|-----------------------|
| P1<br>(n=17) | 0%          | 6%                   | 0%                | 0%                 | 18%               | 0%           | 24%                  | 12%            | 6%          | 0%                    | 0%                | 0%                | 0%              | 18%                   |
| P2<br>(n=45) | 91%         | 31%                  | 64%               | 7%                 | 24%               | 13%          | 13%                  | 2%             | 18%         | 13%                   | 69%               | 7%                | 16%             | 100%                  |
| P3<br>(n=9)  | 100%        | 89%                  | 56%               | 78%                | 100%              | 89%          | 89%                  | 78%            | 22%         | 11%                   | 89%               | 78%               | 22%             | 100%                  |
|              | Finish line | Socratic questioning | Stepwise guidance | No worked examples | Explain reasoning | Use evidence | Compare alternatives | Justify claims | Short turns | One question per turn | Structured format | No direct answers | Respectful tone | High task specificity |

**Figure 1.** Prompt-feature prevalence within each of the three prompt patterns.

## 5. Full Model Details

Model M1 is a linear mixed model for DOK reached with random intercepts for teacher and for activity within teacher, giving teacher variance 0.21 and activity variance 0.22; adding a student random intercept leaves the pattern contrasts null (P2 +0.05, [−0.26, +0.36]; P3 +0.03) with a student standard deviation of 0.27 against 0.38 for activity and 0.37 for teacher. Model M2 is

a logistic regression of goal achievement over DOK 2 and 3 targets with activity-clustered standard errors; the Wald test of the pattern block gives  $p = .65$ . Model M2b restricts M2 to the DOK 3 stratum (Wald  $p = .29$ ). Model M2s uses the  $k = 2$  silhouette-optimal clustering (epistemic-guarded odds ratio 1.45, [0.62, 3.35]). Model M3 re-estimates goal achievement as a generalized linear mixed model with teacher, activity, and student random intercepts, fitted in R; the pattern estimates are P2 odds ratio 0.86 ([0.38, 1.95]) and P3 1.25, with a pattern-block test of  $p = .73$ , and two-way clustering by activity and student leaves the intervals unchanged. Restricting to each student's first conversation gives a pattern-block  $p = .47$ , and across 200 random one-conversation-per-student draws the pattern effect is null in 199 while the target association holds in all 200. For the P3 identification, all nine P3 activities target DOK 3 ( $r = 0.33$  with target, VIF 1.68) and come from five teachers who each also wrote P1 or P2 activities; within teacher the P3 odds ratio is 1.21 ([0.50, 2.94]) and within the DOK 3 stratum 0.73 ([0.23, 2.30]). Model M4 replaces patterns with feature counts; each scaffolding feature is associated with  $-0.18$  DOK levels ( $[-0.31, -0.05]$ ) and achievement odds of 0.68 ([0.48, 0.95]), and each guardrail with  $+0.46$  levels ( $[+0.13, +0.78]$ ) and achievement odds of 3.29 ([1.39, 7.79]). Neither M4 association appears under the conversation-level label, and the within-discipline M4 estimates are unstable, so M4 is reported as exploratory only.

The discrete-time hazard model is a logistic regression on the person-period file of typed student turns before first attainment, with the log typed-turn index, log conversation length, pattern or activity fixed effects, fixed effects for the DOK state of the preceding typed turn, and indicators for the AI's moves on the preceding turn; the fixed-effects variant is estimated within the mixed-outcome activities, and the student-intercept variant is fitted in R. The state term is required because the tutor's move choice tracks the student's state: a Socratic question follows 81% of DOK 2 turns, 68% of DOK 1 turns, and 46% of DOK 0 turns. Once the prior state is held, no move other than the Socratic question shifts the hazard; requests for evidence, requests for an attempt, stepwise guidance, hints, and affirm and extend do not. The interaction of the Socratic estimate with the prior state is null ( $p = .17$ ; within-state estimates 1.71, 1.84, and 1.73, each interval crossing one). Table 2 reports the Socratic estimate across specifications. Full coefficient files for every model accompany the analysis code.

**Table 2.** Socratic-question hazard odds ratio across specifications, DOK 3 stratum, corrected typed-turn risk set. Unless marked unconditioned, the specification conditions on the preceding typed turn's DOK state; the primary specification (L1) adds activity fixed effects and the log-length control within mixed-outcome activities.

| Specification                                    | Odds ratio [95% CI] |
|--------------------------------------------------|---------------------|
| L1, activity FE, log length (primary)            | 1.87 [1.13, 3.10]   |
| L3, no log length                                | 2.03 [1.22, 3.36]   |
| L2, + highest DOK reached so far                 | 1.83 [1.10, 3.03]   |
| GLMM with student random intercept               | 1.88 [1.29, 2.75]   |
| L4, pattern covariates                           | 1.88 [1.20, 2.95]   |
| Transition model, all turns                      | 2.19 [1.56, 3.08]   |
| Activity FE, log length, no state term           | 1.81 [1.04, 3.13]   |
| Fixed five-turn horizon (unconditioned)          | 2.26 [1.04, 4.92]   |
| Conversation-level outcome label (unconditioned) | 1.90 [0.93, 3.89]   |
| First conversation per student (unconditioned)   | 1.96 [0.95, 4.04]   |
| ELA (unconditioned)                              | 2.90 [1.05, 8.02]   |
| STEM/CTE (unconditioned)                         | 1.58 [0.73, 3.45]   |

**Persistence classification.** Among the 349 achieved DOK 3 conversations, 229 (66%) fall below target on at least one post-attainment typed turn. The first drop is a lower-level substantive answer under continued tutor press in 124 conversations, a non-substantive answer under continued press in 50, and a task-driven step, a closing, procedural, or submission turn, in 55; task-driven drops rise to 37% of first drops in the final fifth of the conversation.

## 6. Validation Details

**Conversation-level DOK.** A stratified random sample of 158 conversations (11.6%), oversampling DOK 2/3 boundary cases, was independently coded by a trained researcher blind to the LLM codes and the target label. Agreement was 93.7% (Cohen's  $\kappa = 0.901$ ; quadratic weighted  $\kappa = 0.908$ ), with  $\kappa = 0.931$  at the DOK 2 versus DOK 3 boundary. The ten disagreements split five LLM-higher and five LLM-lower, showing no directional bias, and did not cluster by subject or target.

**Turn-level codes.** Two trained researchers (HC1, HC2) independently coded the full stratified sample of 136 conversations (1,585 student turns and 1,597 AI turns), blind to the LLM codes and to each other. Table 3 reports LLM agreement with HC1 per code, together with human-human reliability and LLM agreement with HC2. On the AI-move codes HC1 and HC2 agree at  $\kappa = 0.79$  to 0.92 (median 0.85).

**Table 3.** Human validation of turn-level codes on the stratified sample of 136 conversations. Unless marked otherwise, rows compare the LLM coder with HC1. Prevalence gives each code's share of coded turns in the primary sample;  $n$  gives the validation-sample turns carrying the coder's code; per-level student-DOK agreement is conditioned on the coder's label.

| Code                                                  | Prevalence | $n$   | Agreement | $\kappa$ |
|-------------------------------------------------------|------------|-------|-----------|----------|
| <i>Student-turn DOK (LLM vs. HC1)</i>                 |            |       |           |          |
| DOK 0 (non-substantive)                               | 39.0%      | 560   | 95.4%     | —        |
| DOK 1 (recall)                                        | 32.2%      | 517   | 85.7%     | —        |
| DOK 2 (skills/concepts)                               | 20.1%      | 380   | 72.6%     | —        |
| DOK 3 (strategic thinking)                            | 8.7%       | 128   | 39.8%     | —        |
| Overall (exact)                                       | —          | 1,585 | 82.3%     | 0.74     |
| Overall (quadratic weighted)                          | —          | 1,585 | —         | 0.89     |
| DOK 2 vs. 3 (both coders in {2, 3})                   | —          | 406   | 80.5%     | 0.46     |
| DOK $\geq 3$ indicator                                | —          | 1,585 | 94.8%     | 0.54     |
| At-or-above-target indicator                          | —          | 1,585 | 92.5%     | 0.77     |
| Goal achieved (conversation level)                    | —          | 136   | 86.0%     | 0.73     |
| <i>Second coder and human-human reliability</i>       |            |       |           |          |
| Student-turn DOK, LLM vs. HC2                         | —          | 1,585 | —         | 0.72     |
| Student-turn DOK, LLM vs. both-agree turns            | —          | 1,463 | 84.8%     | 0.78     |
| Student-turn DOK, HC1 vs. HC2 (exact)                 | —          | 1,585 | 91.0%     | 0.87     |
| Student-turn DOK, HC1 vs. HC2 (quad. weighted)        | —          | 1,585 | —         | 0.94     |
| DOK 2 vs. 3 boundary, HC1 vs. HC2                     | —          | —     | —         | 0.92     |
| At-or-above target, HC1 vs. HC2                       | —          | 1,585 | —         | 0.93     |
| Goal achieved, HC1 vs. HC2                            | —          | 136   | —         | 0.93     |
| <i>AI-move codes (share of AI turns; LLM vs. HC1)</i> |            |       |           |          |
| Socratic question                                     | 52.8%      | 724   | 90.9%     | 0.82     |
| Affirm and extend                                     | 34.5%      | 576   | 90.9%     | 0.79     |
| Stepwise guidance                                     | 20.6%      | 357   | 87.7%     | 0.58     |
| Request attempt                                       | 20.3%      | 408   | 89.6%     | 0.73     |
| Hint                                                  | 20.2%      | 335   | 88.5%     | 0.68     |
| Final answer                                          | 18.0%      | 358   | 96.4%     | 0.90     |
| Request evidence                                      | 8.4%       | 108   | 97.1%     | 0.79     |
| Redirect                                              | 7.7%       | 120   | 97.0%     | 0.76     |
| Other                                                 | 5.8%       | 99    | 96.5%     | 0.75     |
| Recap                                                 | 4.3%       | 59    | 97.7%     | 0.68     |
| Closure                                               | 3.1%       | 30    | 99.1%     | 0.80     |

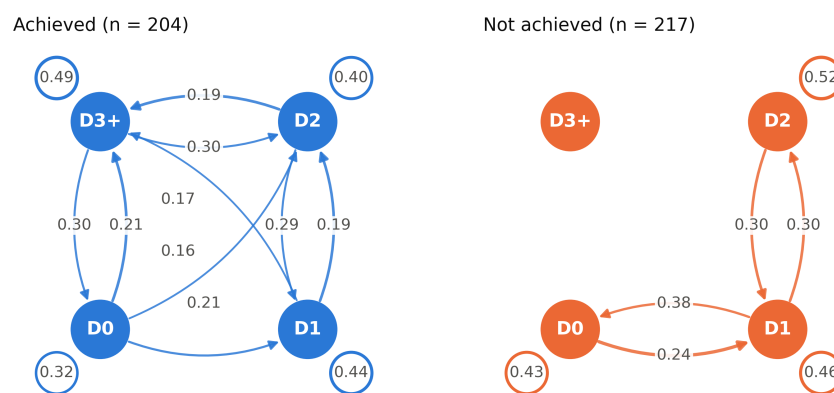
The LLM's disagreements are adjacent levels in over 98% of cases against either human, and the LLM assigns the higher level in 229 of its 285 disagreements with HC1 and 270 of its 308 with HC2, a leniency concentrated at the DOK 2 versus DOK 3 boundary. In the DOK 3 stratum of the validation sample (83 conversations), the achievement rate is 27.7% under HC1, 25.3% under HC2, 43.4% under the LLM turn codes, and 32.5% under the conversation-level label. Gwet's AC1, computed alongside  $\kappa$  for the rare codes, gives the same ordering; the full agreement files accompany the analysis code.

**Turn-level RQ2 quantities under the human codes.** On the 83 human-coded DOK 3 conversations, 23 achieve under HC1, 21 under HC2, and 36 under the LLM. The median first at-target typed turn is 3 under both humans (IQR 2–7 under HC1, 2–6 under HC2) against 2.5 under the LLM, and by typed turn 4 the humans place 61% and 62% of attainments against 72% for the LLM. After attainment, 22% and 24% of human-coded achieved conversations sustain the target to the end against 19% under the LLM, 61% and 57% fall below target at least once against 67%, and 22% and 24% regress without returning against 8%, so the LLM's leniency mainly inflates returns to target rather than attainments. The two humans select the same first-attainment turn in 19 of the 21 conversations achieved under both, and the same persistence class in 19.

**Agreement between coding layers.** The turn-level and conversation-level DOK labels agree on 85.7% of primary-sample conversations. In 134 conversations the turn coder locates an at-target turn the conversation label does not credit, 62% of them containing exactly one such turn, and in 39 conversations the conversation-level label is the higher of the two. Disagreement is spread across subjects. In the excluded World Language activities the two coding layers disagreed on half of the conversations, concentrated on whether submitted creative-writing drafts constitute strategic thinking.

**Table 4.** Key quantities by discipline. ELA is 656 conversations, 38 activities, 5 teachers; STEM/CTE is 557 conversations, 33 activities, 10 teachers. Hazard odds ratios use the corrected typed-turn risk set with prior-state conditioning; within-activity move differences use the length-adjusted specification of main-text Table 4.

| Quantity                                | ELA                   | STEM/CTE                                    |
|-----------------------------------------|-----------------------|---------------------------------------------|
| DOK 3 achievement                       | 48% ( $n = 421$ )     | 45% ( $n = 320$ )                           |
| Mean DOK reached (DOK 3 targets)        | 2.21                  | 2.04                                        |
| DOK 2 achievement                       | 68%                   | 71%                                         |
| Median first at-target typed turn       | 2                     | 3                                           |
| Socratic-question hazard OR             | 2.9 [1.1, 8.0]        | 1.6 [0.7, 3.5]                              |
| Sustained after attainment              | 20%                   | 10%                                         |
| Ends on an at-target turn               | 53%                   | 33%                                         |
| Affirm-and-extend within-activity diff. | +11.0 pp [4.4, 17.6]  | +22.8 pp [17.2, 28.3]                       |
| Socratic-question within-activity diff. | +4.7 pp ( $p = .14$ ) | +12.7 pp ( $p < .001$ )                     |
| Missing transition in not-achieved      | DOK 2 $\rightarrow$ 3 | DOK 1 $\rightarrow$ 2 and 2 $\rightarrow$ 3 |

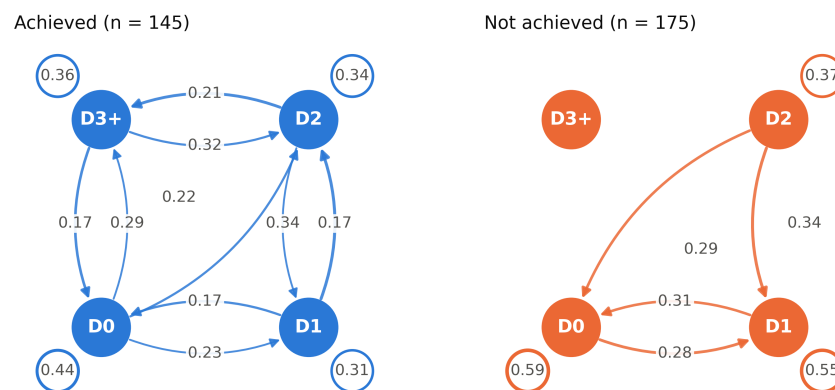


**Figure 2.** Student DOK transition networks in ELA, DOK 3 stratum (204 achieved, 217 not-achieved conversations). Edge labels give transition probabilities after pruning the edges with the lowest 15% of probabilities; DOK 4 is merged into DOK 3+. The failing ELA conversation reaches explanation (DOK 2) and lacks the step to argument.

## 7. Discipline Split

The pattern-by-discipline interaction on achievement is null (Wald  $p = .22$ ), and no pattern contrast appears within either discipline (ELA P2 vs. P1 odds ratio 0.80, [0.31, 2.09]; STEM/CTE 0.81, [0.20, 3.34]). The discipline main effect on achievement is null once pattern and target are controlled (STEM/CTE odds ratio 0.92, [0.47, 1.81]), and the same holds for DOK reached ( $-0.04$  levels,  $[-0.54, +0.45]$ ). Discipline-specific transition network figures and attainment curves accompany the analysis outputs.

Figures 2 and 3 show the discipline-specific transition networks that carry the failure-mode contrast. In ELA the not-achieved network reaches DOK 2 and lacks only the edge into DOK 3+. In STEM/CTE the not-achieved network holds no edge from DOK 1 to DOK 2, so both upward steps are missing.



**Figure 3.** Student DOK transition networks in STEM/CTE, DOK 3 stratum (145 achieved, 175 not-achieved conversations). Edge labels as in Figure 2. The failing STEM/CTE conversation sits in the recall loop (DOK 1 self-loop 0.55) with no edge from recall to explanation, so both upward steps are missing.